



Contact Dermatitis

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Last updated: Apr 2025

Contact dermatitis is inflammation of the skin caused by direct contact with irritants (irritant contact dermatitis) or allergens (allergic contact dermatitis). Symptoms include pruritus and sometimes a burning pain. Skin changes include erythema, scaling, skin swelling, and sometimes blistering and ulceration. The location depends on the site of contact. Diagnosis is by exposure history, examination, and sometimes skin patch testing. Treatment includes topical corticosteroids, antipruritics, and avoidance of irritants and allergens.

(See also [Definition of Dermatitis](#).)

Pathophysiology of Contact Dermatitis

Contact dermatitis is caused by irritants or allergens.

Irritant contact dermatitis (ICD)

ICD is a nonspecific inflammatory reaction to toxic substances contacting the skin. Numerous substances can irritate the skin, including:

- Chemicals (eg, acids, alkalis, solvents, metal salts)
- Soaps (eg, abrasives, detergents)
- Plants (eg, poinsettias, peppers)
- Chronic moisture (eg, from body fluids, urine, and saliva)

Properties of the irritant (eg, extreme pH, solubility in the lipid film on skin), environment (eg, low humidity, high temperature, high friction), and patient (eg, very young or old) influence the likelihood of developing ICD. ICD often can be divided into categories:

- Acute ICD: Potent irritants, such as caustic chemicals, can damage the skin immediately, typically manifesting with acute burning or stinging pain.

- Chronic or cumulative ICD: Less potent irritants require longer (chronic) or repeated (cumulative) periods of skin contact to cause ICD; these forms typically manifest with pruritus.

Occupational ICD is ICD caused by one or more of the many possible work-related skin irritants. It can be acute, chronic, or cumulative.

Atopic disorders increase risk of ICD because of impaired skin barrier function and lower threshold for skin irritation.

Phototoxic dermatitis (see [Chemical photosensitivity](#)) is a variant in which a topical (eg, perfumes, [coal tar](#)) or ingested (eg, psoralens) agent becomes a skin toxin only after exposure to ultraviolet light, most typically long-wave ultraviolet light (UVA). Phototoxic dermatitis, therefore, occurs only in UV-exposed skin, typically with a sharp demarcation.

Allergic contact dermatitis (ACD)

ACD is a type IV, T-cell-mediated, delayed-type hypersensitivity reaction to an environmental allergen that has 2 phases:

- Sensitization to an antigen
- Allergic response after reexposure

In the sensitization phase, allergens are captured by Langerhans cells (dendritic epidermal cells). When activated by innate immunity cascades, these cells migrate to regional lymph nodes, where they process and present the antigen to naive, antigen-specific T cells. When a naive T cell recognizes its antigen via binding to its T-cell receptor, it expands clonally and differentiates into memory/effector T cells. The sensitization phase, which is asymptomatic, may be brief (6 to 10 days for strong sensitizers such as poison ivy) or prolonged (years for weak sensitizers such as sunscreens, fragrances, and glucocorticoids). During differentiation, sensitized T cells become able to express cutaneous homing antigens (eg, cutaneous lymphocyte antigens) that enable them to migrate from cutaneous capillaries to the epidermis. When antigen-presenting cells present the antigen to the sensitized T cells, the T cells can expand and trigger an inflammatory reaction at the location of the skin contact (elicitation phase of ACD), resulting in the characteristic symptoms and signs of ACD.

Multiple allergens can cause ACD (see table [Causes of Allergic Contact Dermatitis](#)). Nickel sulfate is the most common contact allergen in most populations. The *Toxicodendron* species of plant (eg, poison ivy, poison oak, poison sumac) accounts for a large percentage of ACD, including moderate and severe cases. The offending allergen is urushiol.

TABLE	
Causes of Allergic Contact Dermatitis	
Cause	Examples
Chemicals used in shoe or clothing manufacturing	Particularly agents used in leather and rubber processing, tanning agents in

Cause	Examples
	shoes, rubber accelerators and antioxidants in apparel (eg, gloves, shoes, underpants), formaldehyde in durable-press finishes, insecticide spray
Cosmetics	Depilatories, nail polish, deodorant
Dyes	Paraphenylenediamines (hair and textile dyes)
Fragrances	Various compounds Ubiquitous in toiletries, soaps, and scented household products
Industrial agents	Many compounds, including acrylic monomers, epoxy compounds, vat dyes, rubber accelerators, and formaldehyde (in plastics and adhesives)
Ingredients in topical medications	Antibiotics (eg, bacitracin , neomycin) Antihistamines (eg, diphenhydramine) Anesthetics (eg, benzocaine) Antiseptics (eg, thimerosal, hexachlorophene) Glucocorticoids Stabilizers (eg, ethylenediamine and derivatives)
Latex	Latex gloves, condoms, catheters, balloons
Metal compounds	
Nickel	Numerous occupational exposures
Chromates	Personal items (eg, belt buckles, watch buckles, jewelry)
Cobalt	
Mercury	
Plants	Poison ivy, oak, and sumac; primrose; cashew shells; mango peel

Photoallergic contact dermatitis is a variant of ACD, in which a substance becomes sensitizing only after it undergoes structural change triggered by ultraviolet light. Reactions may extend to non-sun-

exposed skin. Typical causes include fragrances (eg, musk ambrette, sandalwood), topical nonsteroidal anti-inflammatory drugs, and sunscreen filters.

Symptoms and Signs of Contact Dermatitis

Irritant contact dermatitis

Acute ICD is more painful than pruritic. Signs range from erythema, scaling, and edema to erosions, crusting, and blistering. Chronic and cumulative ICD is more often pruritic.

Allergic contact dermatitis

ACD is more pruritic than painful. Skin changes range from erythema, scaling, and edema, through vesiculation to severe swelling with bullae. Changes often occur in a pattern, distribution, or combination that suggests a specific exposure, such as linear streaking on an arm or leg (eg, due to brushing against poison ivy) or circumferential erythema (under a wristwatch or waistband). Linear streaks are almost always indicative of an external allergen or irritant.

The site of contact is where the allergen contacted the skin, very often the hands. Although the palms and the palmar sides of the fingers are most exposed, ACD often starts in the web spaces between fingers because the thick stratum corneum on the palms and palmar sides of the fingers (also on the soles) prevents or delays allergen penetration. With airborne exposure (eg, perfume aerosols), areas not covered by clothing are predominantly affected. Although ACD is typically limited to the site of contact, it may later spread because of scratching and autoeczematization (id reaction, a dermatitis at sites remote from the site of the initial inflammatory problem or infection). Because of the time needed to recruit and expand T cells in the epidermis, ACD typically takes ≥ 1 day after exposure to become noticeable and 2 to 3 days to become further aggravated (crescendo reaction). In contrast, ICD typically decreases in intensity [decrecendo reaction] after 1 or 2 days.

Pearls & Pitfalls

- Lesion shape or pattern (linear streaks are almost always indicative of an external allergen or irritant) can help differentiate contact dermatitis from other forms of dermatitis.

Contact Dermatitis (Allergic)



Hide Details

Skin manifestations of allergic contact dermatitis range from erythema through vesiculation to edema with bullae. Changes often occur in a pattern or distribution that suggests a specific exposure. In this image, linear streaking on an extremity suggests plant contact (eg, poison ivy or poison sumac).

Image provided by Thomas Habif, MD.

Contact Dermatitis (Tattoo)



Hide Details

This photo shows allergic contact dermatitis on the forearm of a child after a temporary, black "henna" tattoo was applied.

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Allergic Contact Dermatitis

IMAGE



This photo shows erythema and scaling in the web spaces between fingers. On the hands, allergic contact dermatitis often starts in the web spaces despite the palmar side typically having more direct contact to allergens. This is because the thick stratum corneum on the palmar side often prevents allergen penetration, whereas the stratum corneum in the web spaces is much thinner.

PHOTO COURTESY OF THOMAS RUENGER, MD, PHD.

Diagnosis of Contact Dermatitis

- Clinical evaluation
- Sometimes patch testing

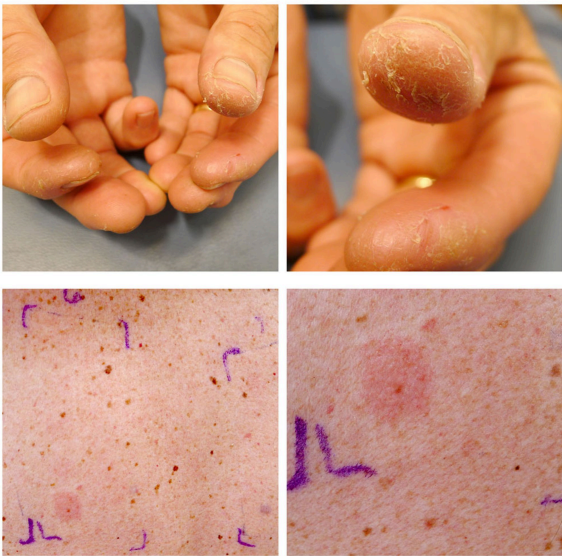
Contact dermatitis can often be diagnosed by skin changes and exposure history. The patient's occupation, hobbies, household duties, recent travel, clothing, topical medication use, and cosmetic use must be considered.

Patch testing is indicated when ACD is suspected and does not respond to treatment, suggesting that the trigger has not been identified. In patch testing, standard contact allergens are applied to the upper back using adhesive-mounted patches containing minute amounts of allergen or plastic chambers containing allergens held in place with porous tape. A standard contact allergen series consists of those allergens shown to be the most common in a particular geographic location (see table [Common Allergens Used in Patch Testing](#)). It can be expanded to include additional substances as indicated (eg, a metal worker series, hairdresser series). The thin-layer rapid-use epicutaneous patch testing (T.R.U.E. TEST) is a simple, easy-to-use kit with 36 of the most common contact allergens that can be applied and interpreted by any health care professional. However, it only detects approximately 50% of clinically relevant contact allergens. If available, patch testing with more extensive test panels is therefore recommended.

Allergen patches typically are applied to the skin on the back, left for 48 hours, and then removed. Skin under the patches is evaluated at 48 hours and at 72 or 96 hours after application for degree of erythema, size of the reaction, swelling, and vesiculation/crusting. A crescendo reaction (increase in the reaction from the first to the second reading) is typical for a positive reaction. False-positive results occur when concentrations provoke an irritant rather than an allergic reaction, and typically cause a decrescendo reaction (decrease in the reaction from the first to the second reading). False-positive results can also occur when reaction to one antigen triggers a nonspecific reaction to others or with cross-reacting antigens. False-negative results occur when patch allergens do not include the offending antigen or when patients have recently undergone immunosuppressive treatment. Definitive diagnosis requires a positive test result and a history of dermatitis in the area where the tested allergen came in contact with the skin.

Patch Testing in Allergic Contact Dermatitis

IMAGE



The top left photo shows erythema, scaling, and lichenification on several of a person's fingertips (close-up, top right). Patch testing showed a strong reaction to 2-hydroxyethyl methacrylate (HEMA) (bottom left; close-up, bottom right). This chemical is contained in a thread-locking glue, which this person handled with his fingertips at work.

PHOTOS COURTESY OF THOMAS RUENGER, MD, PHD.

TABLE

Common Allergens Used in Patch Testing

Agent	Sources
<u>Bacitracin</u>	In topical antibiotic preparations
Balsam of Peru (myroxylon)	A flavoring agent for drinks and tobacco, as well as a fixative and fragrance in perfumes; also occurs in many topical

Agent	Sources
	medications, dental agents, and other products Chief allergens: Esters of cinnamic and benzoic acid, vanillin Cross-reactions with colophony (rosin) and balsam of Tolu, cinnamates, benzoates, styrax, and tincture of benzoin Probably also some phototoxicity
Black rubber mix	In rubber May cross-react with hair dyes
Bronopol	A preservative found in cosmetics, shampoos, and skin care products; also in some detergents and cleaning agents
Budesonide	Screens for class B corticosteroid allergy and should initiate search for sensitization to other topical corticosteroids
Caine mix	Contains 3 topical anesthetics: Benzocaine, dibucaine hydrochloride, and tetracaine hydrochloride Often used in dentistry but also widely found and used in topical preparations to reduce itching, pain, and stinging and widely used in hemorrhoidal preparations and cough syrups
Carba mix	Used as an accelerator in rubber, rubber glues, vinyl, and some pesticides
Cl+ Me- isothiazolinone and methylisothiazolinone	Occur in cosmetics and skin care products, some drugs, household cleaning products, and certain industrial fluids and greases
Cobalt dichloride	Occurs in some paints, cement, metal, and metal-plated objects Coactivity with nickel (which is not cross-

Agent	Sources
	sensitivity)
Colophony (rosin)	Used by string players (violinists are especially prone to rosin allergy), baseball players, and bowlers Derived from several conifer species Occurs in cosmetics, adhesives, lacquers, varnishes, soldering fluxes, paper, and many other industrial products
Diazolidinyl urea	A preservative with broad-spectrum application found in cosmetics, shampoos, skin care products, and cleaning agents
Disperse blue 106	A dark blue textile dye found in fabrics colored dark blue, brown, black, purple, and some greens
Epoxy resin	A low molecular weight (340) epoxy based on bisphenol A and epichlorohydrin Is a sensitizer only when uncured or incompletely cured
Ethylenediamine	Used as an emulsifier and stabilizer in certain topical medications, eye drops, some industrial solvents, curing agents for certain plastics, and anticorrosion agents
Formaldehyde and formaldehyde releasers	Released by quaternium-15, a germicidal agent, and occasionally by imidazolidinyl urea Used widely in formulation of plastics, resins for clothing, glues, and adhesives
Fragrance mixes	Can contain alpha-amyl cinnamic alcohol, cinnamic aldehyde, cinnamic alcohol, oak moss absolute, hydroxycitronellal, eugenol, isoeugenol, geraniol, citral, citronellol, coumarin, farnesol, hexyl cinnamal,

Agent	Sources
	hydroxyisohexyl-3-cyclohexene, and carboxaldehyde Occurs in many toiletries, soaps, aftershave lotions, shampoos, and scented household products and in many industrial products (eg, cutting fluids)
Gold sodium thiosulfate	Found in gold or gold-plated jewelry or dental restorations
Hydrocortisone 17-butyrate	A corticosteroid found in creams or lotions used to treat inflammatory skin diseases; also present in some ear and eye drops
Imidazolidinyl urea	A preservative with broad-spectrum application found in cosmetics, shampoos, skin care products, and cleaning agents
Mercaptobenzothiazole	Occurs in rubber, adhesives, and coolants
Mercapto mix	Occurs in rubber, glues, coolants, and other industrial products
Methyldibromo-glutaronitrile	Found in paints, adhesives, and oils
Neomycin sulfate	Found in topical antibiotics, first-aid creams, ear drops, and nose drops; possible delay (approximately 4–5 days) in patch test reaction (so reading should be done at 7 days when possible)
Nickel sulfate	The most common allergen Used in jewelry, dentures, scissors, razors, eyeglass frames, silverware, and foods (eg, canned foods, foods cooked in nickel utensils)
Paraben mix	Five parabens: Methyl, ethyl, propyl, butyl, and benzyl parahydroxybenzoates, which are the most common preservatives used worldwide and are in

Agent	Sources
	numerous creams and cosmetics and in some industrial oils, fats, and glues
Parthenolide	A sesquiterpene lactone naturally present in the traditional medicine herb, feverfew (<i>Tanacetum parthenium</i>); found in natural medicine and cosmetics Cross-reacts with other genera in <i>Compositae</i> and <i>Magnoliaceae</i>
Potassium dichromate	Occurs in cement (in minute amounts), in tanning solutions for leather, and in safety matches Used in photography, electroplating solutions, many anticorrosives, paints, glues, pigments, and some detergents
<i>p</i> -Phenylenediamine (PPD)	Occurs in hair dyes, some inks, photo developers, and textile dyes
<i>p</i> -Tert-butylphenol formaldehyde resin	A resin formed by condensation between <i>p</i>-tert-butylphenol and formaldehyde Occurs in leather finishes (especially shoes), paper, fabrics, rockwood, furniture, and certain glues
Quaternium	Common preservative occurring in cosmetics and in some household cleaners and polishes
Quinolone mix	Contains clioquinol and chlorquinaldol Antimicrobials occurring in certain medicated creams and ointments, medicated bandages, and veterinary products
Thimerosal	Preservative in contact lens solutions, certain cosmetics, nose and ear drops, and injectables Source often not identified
Thiuram mix	Common rubber allergen

Agent	Sources
	Also occurs in adhesives, certain pesticides, and medications (eg, disulfiram)
Tixocortol-21-pivalate	Screens for class A corticosteroid allergy; found in buccal, nasal, throat, and rectal but not topical corticosteroid preparations
Wool alcohols	Part of lanolin ; found in many cosmetics, ointments, sunscreens, and prescription and over-the-counter topical medications

Treatment of Contact Dermatitis

- Avoidance of offending agents
- Supportive care (eg, cool compresses, dressings, antihistamines)
- Corticosteroids (most often topical but sometimes oral)

Contact dermatitis is prevented by avoiding contact with the skin irritant or allergen; patients with photosensitive contact dermatitis should avoid exposure to the sun (eg, by staying out of the sun or wearing sun-protective clothing) (1).

Topical treatment includes cool compresses (saline or Burow solution) and corticosteroids. Patients with mild to moderate ACD are given mid- to high-potency topical corticosteroids (eg, [triamcinolone](#) 0.1% ointment or [betamethasone](#) valerate cream 0.1%).

Oral corticosteroids, given as a 7 to 14 day course, can be used for severe blistering or extensive disease.

Systemic antihistamines (eg, [hydroxyzine](#), [diphenhydramine](#)) help relieve pruritus; antihistamines with low anticholinergic potency, such as low-sedating H1 blockers, are not as effective.

Wet-to-dry dressings can soothe oozing blisters, dry the skin, and promote healing.

Prognosis for Contact Dermatitis

Resolution may take up to 3 weeks after discontinuation of exposure. Reactivity is usually lifelong, so identified allergens must be avoided lifelong.

Patients with [photoallergic contact dermatitis](#) can have flare-ups for years when exposed to sun (persistent light reaction); however, this is very rare.

Prognosis reference

1. [Fonacier L, Bernstein DI, Pacheco K, et al](#). Contact dermatitis: a practice parameter-update 2015. *J Allergy Clin Immunol Pract*. 2015;3(3 Supl):S1-S39. doi:10.1016/j.jaip.2015.02.009

Key Points

- Contact dermatitis can be caused by irritants (eg, plants, soaps, chemicals, body fluids) or by allergens.
- Symptoms can include predominantly pain (for irritant contact dermatitis) or pruritus (for allergic contact dermatitis).
- Diagnosis is usually clinical.
- Patch testing is helpful when allergic contact dermatitis is suspected and the trigger has not been identified.
- Treatment commonly includes cool compresses, topical corticosteroids, and systemic antihistamines as needed for pruritus.

Drug Information for the Topic



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